Q1) What is quantum entanglement meant? Explain with an example of an entangled two-qubit system.

Quantum entanglement is a phenomenon where **two or more quantum systems become correlated in such a way that the state of one system cannot be described independently of the other**, even if the systems are separated by a large distance

In an entangled system a measurement performed on one subsystem instantaneously influences the state of the other, a link Albert Einstein famously called "spooky action at a distance." Crucially, this correlation is stronger than any possible correlation in classical physics.

Mathematical Definition

A composite quantum state $|\Psi\rangle$ is **entangled** if it cannot be written as a simple **tensor product** (or product state) of the individual states $|\psi_A\rangle$ and $|\phi_B\rangle$:

$$|\psi\rangle_{\text{entangled}} \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

If a state *can* be written as a product state,

$$|\psi\rangle_{\text{separable}} = |\psi_A\rangle \otimes |\psi_B\rangle$$

it is called a **separable state**, meaning the two parts are independent.

Example: Two-Qubit Entangled System

Consider a system of **two qubits**, A and B. One of the simplest entangled states is the **Bell state**:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Here:

- $|00\rangle$ means qubit A is 0 and qubit B is 0
- $|11\rangle$ means qubit A is 1 and qubit B is 1
- The state is a **superposition** of both possibilities.

Key features:

1. **Cannot be separated:** There's no way to write $|\Phi^+\rangle$ as a product of a state of A and a state of B.
2. **Perfect correlation:** If we measure qubit A and find it in state $|0\rangle$, qubit B will also be $|0\rangle$. If A is $|1\rangle$, B is $|1\rangle$.
3. **Instantaneous correlation:** This correlation exists even if A and B are far apart, which is why entanglement is often described as "spooky actic at a distance."

3. The results are **perfectly correlated**, even if the two qubits are separated by large distances — a key feature of entanglement.

Q2) State and explain Bell's Theorem in the context of local hidden variable theories.

Bell's Theorem is a crucial result in quantum mechanics that establishes a fundamental difference between the predictions of quantum theory and any physical theory based on local hidden variables (LHVs).

In simple terms, Bell's Theorem proves that no local theory can reproduce all the statistical correlations predicted by quantum mechanics for entangled particles.

Before Bell's work, many scientists (most famously Einstein) believed that all physical systems must have **definite properties** (like position, spin, etc.) even before measurement. They hypothesized that quantum mechanics was incomplete and that the randomness was only *apparent*

A **Local Hidden Variable Theory** posits two main ideas:

1. **Realism (Hidden Variables):** The results of any quantum measurement are predetermined by **hidden variables** (unseen properties) that the particles carry with them from the moment they are created. There is no true randomness; we just don't know the values of these hidden variables.
2. **Locality:** Physical influences cannot travel faster than the speed of light. Therefore, the measurement performed on Qubit A cannot instantaneously influence the properties or measurement outcome of Qubit B if they are separated.

Bell's Theorem states that if the world were governed by local hidden variable theories (where particles have predetermined properties and no influence travels faster than light), then certain mathematical inequalities—called Bell inequalities—must always be satisfied.

A common form of the Bell inequality is the **CHSH (Clauser, Horne, Shimony, Holt) inequality**, which imposes a bound on the combination of four correlation functions, often written as:

$$|S| \leq 2$$

Where S is the **Bell correlation parameter**.

$$|\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle| \leq 2\,.$$

Physicist John Bell derived these inequalities in 1964. Crucially, he then showed that **quantum mechanics predicts correlations that can violate this inequality.**

When experiments (known as **Bell tests**) were performed on entangled particles, the results consistently matched the predictions of quantum mechanics and **violated the Bell inequality**

Quantum mechanics, specifically for maximally entangled states like the Bell states, predicts that the correlation parameter S can reach a maximum value of:

$$|S|_{\mathrm{QM}} = 2\sqrt{2} \approx 2.828$$

Since $2.828 > 2$, the quantum prediction directly violates the local hidden variable limit.

The Conclusion
Since the experimental results violate the limit imposed by *any* local hidden variable theory, the theorem proves that the universe **cannot** be described by a theory that is both **local and realistic**.

This forces us to accept at least one of the following:

- **Non-Locality:** Physical influences can be instantaneous (the "spooky action" of entanglement is real), **OR**
- **Non-Realism:** Physical properties are not predetermined but are genuinely created only at the moment of measurement.

Q3) What Do You Mean by Purification in Quantum Information Theory?

Purification in quantum information theory is a mathematical technique used to represent a **mixed quantum state** as a **pure state** in a larger, combined Hilbert space.

It states as every **mixed state** (described by a **density matrix**) can be seen as the **partial state** of a **larger system** that is in a **pure state**

Mathematical Definition

Purification is the process by which we create a reference system B such that, given the system A, the state |φAφB > is a pure state.

If ρ_A is a mixed state, we can use purification to analyze the system as a pure state by expanding the Hilbert space to the larger space defined by $|\phi_A \phi_B\rangle$. Suppose

$$\rho_A = \sum_i p_i |a_i\rangle\langle a_i|$$

Let $|b_i\rangle$ be an orthonormal basis for system B. The purification is then

$$|\phi_B\rangle = \sum_i \sqrt{p_i}|a_i\rangle \otimes |b_i\rangle \tag{7.47}$$

So the density matrix of the mixed state ρA. And we take the **partial trace** The state |φB> then |φB> is a purification of ρA if

$$\rho A = \mathrm{Tr}_B(|\varphi B\rangle\langle\varphi B|)$$

4. Example of Purification

Let's say we have a qubit in a mixed state:

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

This means the qubit has a 50% chance of being $|0\rangle$ and 50% chance of being $|1\rangle$.

We can purify it by introducing a second qubit (the environment) and writing:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This is a **pure entangled state** of two qubits.

If we look only at the first qubit and ignore the second one, its state appears mixed — exactly the ρ we started with.

Q4) Describe how measurement on one qubit of an entangled pair affects the other qubit's state.

In quantum mechanics, when two qubits are **entangled**, their states become **interdependent** — meaning that the state of one qubit **cannot be described independently** of the other.
Even if the qubits are separated by a large distance, they still share a single combined (joint) wavefunction.

So, when we **measure one qubit**, we immediately affect — or more precisely, **determine** — the state of the other qubit.

This effect is a direct consequence of **quantum entanglement** and **wavefunction collapse**.

Let's take a simple example: the **Bell state**

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This means:

- Both qubits are in a **superposition** of being 0 and 1.
- The two qubits are **perfectly correlated** — they are always the same (either both 0 or both 1).

Before measurement, neither qubit has a definite value.
Each one is **50% likely** to be measured as 0 or 1, but **their outcomes are linked**.

Suppose we measure the **first qubit** in the computational basis.

- If the first qubit is measured as $|0\rangle$, the **entire entangled state collapses** to $|00\rangle$.
- If the first qubit is measured as $|1\rangle$, the state collapses to $|11\rangle$.

In both cases, the **second qubit's state becomes fixed** — it instantly becomes $|0\rangle$ if the first qubit was 0, or $|1\rangle$ if the first qubit was 1.

Thus, measuring one qubit **determines** the other's state, even if no direct interaction occurs between them.

Q5) Derive the Bell inequality and show how quantum mechanics violates it.

Imagine two distant observers, **Alice** and **Bob**. Each chooses one of two measurement settings:

- Alice chooses a or a'.
- Bob chooses b or b'.

Each measurement returns a result A or B respectively, with values ± 1.
In a local hidden-variable model, outcomes are determined by a shared hidden parameter λ and local response functions:

$$A(a,\lambda) \in \{+1,-1\}, \qquad B(b,\lambda) \in \{+1,-1\}.$$

Outcomes may vary with λ; experimental statistics are averages over a probability distribution $\rho(\lambda)$ (with $\rho(\lambda) \geq 0$ and $\int d\lambda\, \rho(\lambda) = 1$).

Define the correlation function:

$$E(a,b) = \int d\lambda\, \rho(\lambda)\, A(a,\lambda)\, B(b,\lambda).$$

We will derive an inequality that any such LHV model must satisfy.

2. Algebraic identity used in the CHSH derivation

Consider, for a fixed λ, the combination

$$S(\lambda) \equiv A(a,\lambda)\,B(b,\lambda)\ +\ A(a,\lambda)\,B(b',\lambda)\ +\ A(a',\lambda)\,B(b,\lambda)\ -\ A(a',\lambda)\,B(b',\lambda).$$

We want to find the bounds on $S(\lambda)$. Factor $S(\lambda)$ as:

$$S(\lambda) = A(a,\lambda)\big[B(b,\lambda) + B(b',\lambda)\big]\ +\ A(a',\lambda)\big[B(b,\lambda) - B(b',\lambda)\big].$$

Because each A and B equals ± 1, examine the possible values of the bracketed terms:

- $B(b,\lambda) + B(b',\lambda)$ can be $+2, 0,$ or $-2.$
- $B(b,\lambda) - B(b',\lambda)$ can be $+2, 0,$ or $-2.$

But note that for any fixed λ, either $B(b,\lambda) + B(b',\lambda)$ equals 0 and $B(b,\lambda) - B(b',\lambda) = \pm 2$, or vice versa, or both are ± 2 and 0 respectively. The key simple bound follows:

Since $A(a,\lambda)$ and $A(a',\lambda)$ are each ± 1,

$$|S(\lambda)| \le 2.$$

4. Quantum-mechanical prediction for an entangled state

Now compute the quantum prediction for a specific entangled state and measurement settings. Use the **singlet state** (one of the Bell states):

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\big(|01\rangle - |10\rangle\big).$$

For spin-1/2 particles (or polarization qubits), quantum mechanics predicts correlation for spin measurements along directions $\hat{a}$ and $\hat{b}$ given by:

$$E_{\text{QM}}(\hat{a},\hat{b}) = -\cos\theta_{ab},$$

where θ_{ab} is the angle between directions $\hat{a}$ and $\hat{b}$. (This is the standard result for the singlet: perfect anti-correlation when $\hat{a} = \hat{b}$.)

We will pick measurement directions (angles) that maximize the violation. A standard choice is:

$$\begin{aligned} a &= 0, \quad & a' &= \tfrac{\pi}{2}, \\ b &= \tfrac{\pi}{4}, \quad & b' &= -\tfrac{\pi}{4}, \end{aligned}$$

Compute the four correlations (θ = difference of angles):

1. $E(a,b) = E(0, \frac{\pi}{4}) = -\cos(\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.
2. $E(a,b') = E(0, -\frac{\pi}{4}) = -\cos(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.
3. $E(a',b) = E(\frac{\pi}{2}, \frac{\pi}{4}) = -\cos(\frac{\pi}{2} - \frac{\pi}{4}) = -\cos(\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.
4. $E(a',b') = E(\frac{\pi}{2}, -\frac{\pi}{4}) = -\cos(\frac{\pi}{2} + \frac{\pi}{4}) = -\cos(\frac{3\pi}{4}) = -\left(-\frac{\sqrt{2}}{2}\right) = +\frac{\sqrt{2}}{2}$.

Now form the CHSH combination

$$S_{\text{QM}} = E(a,b) + E(a,b') + E(a',b) - E(a',b').$$

Form the CHSH combination:

$$S_{\text{QM}} = E(a,b) + E(a,b') + E(a',b) - E(a',b') = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -2\sqrt{2}.$$

Hence

$$|S_{\text{QM}}| = 2\sqrt{2} \approx 2.828 > 2.$$

So quantum mechanics predicts a value of $|S|$ up to $2\sqrt{2}$ (Tsirelson bound), which **violates** the Bell (CHSH) bound of 2 allowed by any local hidden-variable theory.

Q6 Describe the process of constructing a pure entangled state that purifies a given mixed state.

Purification in quantum information theory is a mathematical technique used to represent a **mixed quantum state** as a **pure state** in a larger, combined Hilbert space.

It states as every **mixed state** (described by a **density matrix**) can be seen as the **partial state** of a **larger system** that is in a **pure state**

Mathematical Definition

A mixed state is represented by a **density matrix**:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where $\{|\psi_i\rangle\}$ are pure states and $\{p_i\}$ are probabilities.

Now follow these steps:

Step 1: Introduce an Ancilla System

Create a new Hilbert space $\mathcal{H}_A$ for the ancilla (environment).
Let it have orthonormal basis states $\{|i\rangle_A\}$, one for each term in the mixture.

Step 2: Form the Combined (Composite) System

Take the **tensor product** of the system space $\mathcal{H}_S$ and the ancilla space $\mathcal{H}_A$:

$$\mathcal{H}_{SA} = \mathcal{H}_S \otimes \mathcal{H}_A$$

Step 3: Construct the Pure Entangled State

Define a pure state in this combined space as:

$$|\Psi\rangle = \sum_i \sqrt{p_i}\, |\psi_i\rangle_S \otimes |i\rangle_A$$

Step 3: Construct the Pure Entangled State

Define a pure state in this combined space as:

$$|\Psi\rangle = \sum_i \sqrt{p_i}\, |\psi_i\rangle_S \otimes |i\rangle_A$$

This state is **entangled** between the system and the ancilla because the two parts cannot be separated into independent states.

Step 4: Verify the Purification

Compute the **density matrix** of the full system:

$$|\Psi\rangle\langle\Psi| = \sum_{i,j} \sqrt{p_i p_j}\, |\psi_i\rangle\langle\psi_j| \otimes |i\rangle\langle j|$$

Now take the **partial trace** over the ancilla (trace out A):

$$\mathrm{Tr}_A(|\Psi\rangle\langle\Psi|) = \sum_i p_i |\psi_i\rangle\langle\psi_i| = \rho$$

5. Example

Suppose you have a mixed qubit state:

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

This represents a qubit that is randomly 0 or 1 with equal probability — a **maximally mixed state**.

To purify it, add another qubit (the ancilla) and construct:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$$

This $|\Psi\rangle$ is a **pure entangled state** (a Bell state).
If we trace out one qubit, the other qubit's state becomes exactly the mixed state ρ above.

Thus, $|\Psi\rangle$ is a **purification** of ρ.

Q7 Explain why entanglement is considered a non-classical correlation between quantum systems.

Entanglement is considered a **non-classical correlation** because it cannot be explained by any **classical theory of probability** or by assuming that particles have **pre-existing, local properties** as in local hidden variable models.

In classical physics, correlations between two systems arise only when both share some common cause or information in the past. In contrast, **quantum entanglement produces correlations that go beyond what classical physics allows** — these correlations violate **Bell inequalities**, proving they have no classical counterpart.

The key differences between classical correlation and quantum Entanglement

Feature	Classical Correlation	Correlation)
Nature of correlation	Arises from shared history or pre-set variables (e.g., both coins made the same way).	Arises from shared quantum state — the systems are described by one **joint wavefunction.**
Mathematical form	Can be described by a **joint probability distribution** over local hidden variables.	Cannot be represented by any local hidden variable model.
Measurement results	Determined by pre-existing properties.	Determined only at the moment of measurement.
Locality	Each system behaves independently; no instant effect between distant systems.	Measurement on one system **instantly affects** the state of the other (nonlocal correlation).
Correlation limit	Must satisfy **Bell's inequality** (≤ 2).	Can **violate Bell's inequality** (up to $2\sqrt{2}$).
Example analogy	Two synchronized watches show the same time because they were set identically.	Two entangled particles show correlated outcomes even if far apart and not causally connected.

Or

Key Reasons Why Entanglement is Non-Classical

1. **No pre-existing definite properties:**
    - In classical systems, measurement reveals a property that already exists.
    - In entangled systems, measurement outcomes are not predetermined; they arise **only when measured**, yet the results on both sides are perfectly correlated.
2. **Instantaneous correlations (non-locality):**
    - Measuring one particle **instantly determines** the outcome of the other, even if they are far apart.
    - This happens **without any signal** traveling between them — it reflects a fundamental quantum link, not classical communication.
3. **Violation of Bell inequalities:**
    - Quantum entangled states produce statistical results that **exceed the bounds** set by classical local hidden variable theories.
    - Experiments confirm that nature follows **quantum predictions**, not classical ones.
4. **No classical probability model can reproduce it:**
    - In classical probability, joint outcomes can always be written as mixtures of independent states.
    - Entangled states **cannot** be written as mixtures of product states — they are **inseparable**.

Q8 Explain the role of Bell's Theorem in establishing the concept of quantum nonlocality

Isme pehle bell's theorem samjana hai jiase pehle kya ho raha tha bell ne kya kiya aur use kya hua

and last me yeh likhna hai

Bell's Theorem is the primary theoretical tool that establishes and confirms the concept of **quantum nonlocality**. It achieves this by providing a definitive test that rules out all competing theories based on the classical principle of locality.

The theorem's role is not to *prove* nonlocality (which is inherent in the quantum formalism), but to *force its acceptance* by showing that the alternative—Local Hidden Variable theories—is mathematically incompatible with experimental results.

Ruling Out Local Realism

Bell's Theorem created a sharp, testable boundary between quantum mechanics and any theory based on **Local Realism** (or Local Hidden Variables, LHVT).[1]

- **Local Realism:** This classical view assumes that particles possess definite properties (Realism) and that influences cannot travel faster than light (Locality).
- **Bell's Inequality:** Bell mathematically proved that *any* theory obeying local realism must satisfy a certain statistical limit, the **Bell Inequality** (e.g., [2]$|S| \le 2$).[3]

By showing this theoretical ceiling, Bell transformed an abstract philosophical debate into a concrete experimental challenge

The Experimental Violation

The crucial step was the experimental testing of entangled quantum systems. Experiments (like those involving photons or electrons) measured the correlations between separated particles and found that they **violate** the Bell Inequality, reaching values up to the quantum limit of $2\sqrt{2} \sim 2.828$

Since the experiments confirmed the quantum prediction and violated the limit of 2:

$$2\sqrt{2} > 2$$

Quantum nonlocality means that:

- The outcome of a measurement on one part of an entangled system can depend on a measurement performed on a distant part, **without any signal traveling between them**.
- This connection happens **instantaneously**, regardless of distance.

Bell's Theorem **proved mathematically** that:

- These nonlocal correlations cannot be explained by classical physics or any local hidden variables.
- Therefore, the quantum world is fundamentally **nonlocal** — although it still respects **causality** (no faster-than-light communication).

So, Bell's Theorem gives **direct evidence** that quantum mechanics involves **real, physical nonlocal correlations** between entangled particles.

Q9 Explain the relationship between mixed states, density matrices, and purification.

Relationship Between Mixed States, Density Matrices, and Purification

These three concepts are closely related — they describe different but connected ways of understanding **quantum systems that are not in pure states.**

---

1 Mixed States

- A **mixed state** represents a **statistical mixture** of several possible pure quantum states.
- It arises when we have **partial information** about a system or when the system is **entangled** with an environment and we only look at part of it.
- Mathematically, a mixed state cannot be described by a single wavefunction $|\psi\rangle$; instead, it's described by a **probability distribution** over several pure states:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where p_i are classical probabilities ($\sum_i p_i = 1$).

2 Density Matrices

- The **density matrix (density operator)** is the mathematical tool used to describe **both pure and mixed states.**
- For a **pure state,** $\rho = |\psi\rangle\langle\psi|$.
- For a **mixed state,** $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$.
- The density matrix allows you to compute **expectation values, probabilities,** and **entanglement** properties without needing to know an exact wavefunction.

✅ **Key properties:**

$$\rho = \rho^\dagger, \quad \rho \geq 0, \quad \mathrm{Tr}(\rho) = 1$$

3 Purification

- **Purification** connects mixed states and pure states.
- It shows that **any mixed state can be viewed as part of a larger pure entangled state** in an extended system.
- Suppose system A is in a mixed state ρ_A. We can always find another system R such that together they are in a **pure state** $|\Psi_{AR}\rangle$, and:

$$\rho_A = \mathrm{Tr}_R(|\Psi_{AR}\rangle\langle\Psi_{AR}|)$$

This process is called **purification**, and the system R is called the **reference (or ancilla)** system.

4 Relationship Summary

Concept	Description	Connection
Mixed State	Statistical combination of several pure states.	Described mathematically by a **density matrix.**
Density Matrix	Operator that represents both pure and mixed states.	Used to calculate all observable quantities of the system.
Purification	Representation of a mixed state as part of a larger pure state.	Shows that every mixed state can come from tracing out part of a **pure entangled system.**

Q10) Discuss the significance of entanglement in quantum communication and quantum teleportation.

Quantum entanglement is a phenomenon where two or more quantum systems become **linked** so that the state of one system instantly determines the state of the other, even if they are far apart.

Mathematically, their combined state **cannot** be written as a simple product of individual states:

$$|\Psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\phi_B\rangle$$

Entanglement in Quantum Communication

Quantum communication aims to transmit **information** using quantum states.
Entanglement provides a way to **correlate particles across distance** in a way that is impossible classically.

Key Roles:

(a) Quantum Key Distribution (QKD)

- Entanglement is used to create **shared, secret correlations** between two distant users (e.g., Alice and Bob).
- In the **E91 protocol** (based on Bell's Theorem), Alice and Bob share entangled photon pairs.
- Measuring their qubits in specific bases produces correlated results that **cannot be copied or intercepted** without disturbing the system.
- Any eavesdropper changes the correlations — revealing their presence.

Entanglement ensures **secure communication** based on the laws of quantum mechanics, not computational hardness.

(b) Superdense Coding

Using entanglement, two parties can **send two classical bits** of information using **only one qubit**.

Alice and Bob share an entangled pair, e.g.:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

By applying one of four local operations (I, X, Y, Z) to her qubit, Alice changes the shared entangled state into one of four possible Bell states.

She then sends **her qubit** to Bob.

Bob performs a **joint (Bell) measurement** on both qubits to decode the message.

Entanglement enables **faster and more efficient communication**, transmitting **two bits of classical information** with just **one qubit**.

(c) Quantum Repeaters and Networks

- Entanglement is essential for building long-distance **quantum networks**.
- It allows **entanglement swapping**, where two distant qubits become entangled even though they never interacted directly.
- This technique is the backbone of the **quantum internet** and **entanglement-based networking**.

Entanglement enables **scalable, long-distance quantum communication** without loss of quantum information.

3. Entanglement in Quantum Teleportation

What is Quantum Teleportation?

Quantum teleportation is a process that allows one party (Alice) to transmit an **unknown quantum state** to another (Bob) **without physically sending the particle** itself.

It works by using **entanglement** and **classical communication** together.

Save as

1. **Entanglement Sharing:**
   Alice and Bob share an entangled pair of qubits, e.g.
$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
2. **Unknown State:**
   Alice has a third qubit in an unknown state:
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$
3. **Joint Measurement:**
   Alice performs a **Bell-state measurement** on her two qubits (the unknown one and her part of the entangled pair).
   This collapses the combined system and projects Bob's qubit into a state related to $|\psi\rangle$.
4. **Classical Communication:**
   Alice sends her **two-bit measurement result** to Bob using a classical channel.
5. **State Reconstruction:**
   Depending on the result, Bob applies a specific **unitary operation** (I, X, Y, or Z) to his qubit.
   After this correction, his qubit becomes an **exact copy** of Alice's original $|\psi\rangle$.

The quantum state $|\psi\rangle$ has been **transferred** (teleported) to Bob — even though the physical particle was never sent.

Entanglement provides the **quantum link** that allows the state information to be transferred. Without entanglement, **teleportation would be impossible** — classical communication alone cannot transmit quantum information.

Q11) Illustrate a Bell-type experiment and describe how measurement outcomes demonstrate entanglement.

A **Bell-type experiment** is designed to test whether the correlations between two entangled particles can be explained by **classical local hidden variable theories**, or if they truly display **quantum entanglement** and **nonlocality.**

Imagine two experimenters — **Alice** and **Bob** — located far apart.
They share a pair of **entangled particles**, such as photons, prepared in a Bell state:
$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This means:

- If Alice measures her photon and gets 0, Bob's photon will also be 0.
- If Alice gets 1, Bob will also get 1.
- But before measurement, neither particle has a definite state — they exist in a **superposition**.

Alice randomly chooses between two measurement settings a and a'. **Bob** randomly chooses between two measurement settings b and b'. Each measurement can result in either **+1** or **−1**.

They repeat this many times to record joint outcomes for all combinations of measurement settings.

If the results were determined by **hidden local properties** (as in classical physics), the **correlations** between Alice's and Bob's outcomes would always satisfy **Bell's inequality**:

$$|S| = |E(a,b) + E(a,b') + E(a',b) - E(a',b')| \leq 2$$

where $E(a,b)$ represents the average correlation between their results for given measurement settings.

Quantum mechanics predicts stronger correlations for entangled particles:

$$E(a,b) = -\cos(\theta_{ab})$$

where θ_{ab} is the angle between the measurement directions.

With the right choice of measurement angles (for example, 0°, 45°, 90°, and –45°), the **quantum prediction** gives:

$$|S| = 2\sqrt{2} > 2$$

This violates Bell's inequality — something **no classical local model can achieve.**

In real Bell-type experiments (such as those using entangled photons):

- Alice and Bob record measurement outcomes that show correlations exactly matching quantum mechanical predictions.
- The **violation of Bell's inequality** confirms that these correlations **cannot** be explained by any **local hidden variable theory**.

Q12) Discuss the importance of purification in quantum error correction and secure communication.

Purification is a fundamental concept in quantum information theory that connects **mixed states** and **pure states**.
It means expressing a **mixed quantum state** (which represents incomplete information) as part of a **larger pure entangled system**.
In simple terms, purification shows that any noisy or uncertain state can be seen as a pure state of a bigger system that includes the environment.

Role in Quantum Error Correction

In quantum systems, **errors occur** when qubits interact with their environment and lose coherence — effectively turning a pure state into a mixed state.

- **Purification provides the theoretical basis** for how to recover the original quantum information.
- It allows us to treat noise as a process of **entanglement with the environment** and to design operations that **reverse or correct** that entanglement.

How purification helps:

- Quantum error correction codes aim to **restore the original pure state** by adding **ancilla qubits** that act like the reference system in purification.
- The encoding process keeps the **total system (data + ancillas)** in a **pure entangled state**, so even if part of it suffers errors, the global purity ensures recovery is possible.

- By performing syndrome measurements and corrections, we can **disentangle** the environment and **re-purify** the logical qubit.

Importance in Secure Quantum Communication:

Purification is also extremely important in **quantum cryptography** and **secure communication** protocols.

1. **Guaranteeing Security through Entanglement:**
   - In protocols like **Quantum Key Distribution (QKD)**, purification provides a way to describe the joint system of Alice, Bob, and any possible eavesdropper (Eve).
   - If the shared state between Alice and Bob is pure (purified), it guarantees that Eve has **no information** about their secret key.
2. **Purification-Based Security Proofs:**
   - Many modern QKD security proofs (like in the **Ekert protocol**) are based on purification arguments — they show that if an eavesdropper tries to gain information, the state becomes mixed, and this disturbance can be detected.
3. **Entanglement Distillation:**
   - Purification techniques are used to **distill high-quality entangled pairs** from noisy ones.
   - This process increases the fidelity of quantum channels, making long-distance quantum communication and teleportation more reliable.

In short, in **quantum error correction**, it helps model and reverse noise caused by environmental entanglement and in **secure quantum communication**, it provides a theoretical basis for proving security and detecting eavesdropping.

Q13 Distinguish between separable and entangled states with suitable mathematical expressions

Aspect	Separable States	Entangled States
Basic Meaning	A separable state is one where the total system's state can be written as the **product** (or mixture) of individual subsystem states.	An entangled state is one where the total system's state **cannot** be written as a simple product of subsystem states.
Mathematical Expression (Pure State)	For two subsystems A and B: [	\Psi_{AB}\rangle =
Mathematical Expression (Mixed State)	A mixed separable state can be written as: $\rho_{AB} = \sum_i p_i \rho_A^i \otimes \rho_B^i$ where p_i are probabilities.	An entangled mixed state **cannot** be expressed in the above form; it has correlations that cannot be separated into local density matrices.
Correlation Type	Only **classical correlations** exist between subsystems.	Shows **quantum correlations** that go beyond any classical explanation.

State of Subsystems	Each subsystem has a **well-defined individual state.**	Each subsystem's individual state is **undefined** — only the joint state is known.
Measurement Effect	Measuring one subsystem **does not affect** the other subsystem.	Measuring one subsystem **instantly determines** the state of the other (nonlocal correlation).
Mathematical Property	The total density matrix equals the **tensor product** of the individual density matrices: $\rho_{AB} = \rho_A \otimes \rho_B$	The total density matrix **cannot** be factored into separate parts.
Quantum Correlation Strength	Correlations can be explained by **local hidden variables** (classical probability).	Correlations **violate Bell's inequalities,** proving they are nonlocal and quantum.
Physical Meaning	Represents **independent systems** with no shared quantum information.	Represents **quantumly linked systems** where states depend on each other.
Use in Quantum Information	Used for **classical mixtures** or **uncorrelated qubits.**	Fundamental for **quantum computing, teleportation, and cryptography.**

Example:

- "Separable → $|01\rangle = |0\rangle_A \otimes |1\rangle_B$"
- "Entangled → $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$"

Q14 When is a quantum state considered entangled? Explain using the concept of separability.

A **quantum state** is said to be **entangled** when the states of two or more subsystems are linked in such a way that the state of each subsystem **cannot be described independently** of the others.
In other words, the total system has to be treated as a single combined entity, and the state cannot be written as a simple product of individual states.

The Concept of Separability

The property of **separability** defines the limit of classical correlation in quantum mechanics. A composite quantum state is separable if and only if all the correlations between its parts can be fully explained by **classical probabilities** over the independent states of the individual systems.

1. Separable States (Not Entangled)

A state of a composite system (A and B) is considered **separable** if it can be written as a simple tensor product or a classical mixture of such products.

This means the systems are **independent** — the measurement on one does not affect the other.

- **Pure Separable State:** The state is a product of the individual state vectors:

$$|\mathbf{\Psi_{AB}}\rangle = |\psi_\mathbf{A}\rangle \otimes |\phi_\mathbf{B}\rangle$$

This means system A is in state $|\psi_A\rangle$ and system B is in state $|\phi_B\rangle$, and they are independent.

- **Mixed Separable State:** The state is a classical probability distribution (a mixture) of independent states:

$$\rho_\mathbf{AB} = \sum_\mathbf{k} \mathbf{p_k}\, \rho_\mathbf{A}^\mathbf{k} \otimes \rho_\mathbf{B}^\mathbf{k}$$

Where ρ_A^k and ρ_B^k are local density operators and p_k is a classical probability.

If a state is separable, all correlations are **classical**.

2. Entangled States (Non-Separable)

A quantum state is **entangled** if it is **not separable**.

- **Definition:** An entangled state is one that **cannot** be written in the separable form shown above.
- Mathematical Condition: For a pure state, entanglement means the state vector cannot be factored into a simple tensor product:

That is:

$$|\psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

or equivalently, for mixed states,

$$\rho_{AB} \neq \sum_i p_i\, \rho_A^{(i)} \otimes \rho_B^{(i)}$$

In such cases, the two subsystems are **not independent** — their quantum properties are intertwined, and measurement of one instantly determines the state of the other. The correlation exhibited by the state is non-classical and cannot be explained by local hidden variables, as demonstrated by Bell's Theorem.

4. Example of an Entangled State:

Consider the **Bell state**:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

5. Example of a Separable (Non-Entangled) State:

$$|\psi\rangle = |0\rangle_A \otimes |1\rangle_B = |01\rangle$$

Q15) Describe how the density matrix representation can be used to identify whether a mixed state is entangled.

The **density matrix** (also called the **density operator**) is a mathematical tool used to describe both **pure** and **mixed** quantum states.
For a mixed state, it expresses the probabilities of the system being in various pure states:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where p_i are probabilities that sum to 1.
This representation is especially important for **entangled systems**, where subsystems cannot always be described independently.

To determine whether a **mixed state** is **entangled** or **separable**, we analyze the **structure** of its density matrix.

- A **separable (non-entangled) mixed state** can be written as:

  $$\rho_{AB} = \sum_i p_i \, \rho_A^{(i)} \otimes \rho_B^{(i)}$$

  This means the total state can be expressed as a **mixture of product states**, showing only classical correlations.
- An **entangled mixed state cannot** be written in that separable form.
  Its density matrix has quantum correlations that cannot be explained by any combination of separable (product) states.

Here are the key ways to check whether a mixed state is entangled, using its density matrix representation:

(a) Check for Non-Separability:
If the density matrix **cannot** be written as a sum of tensor products of individual subsystem density matrices,

$$\rho_{AB} \neq \sum_i p_i \, \rho_A^{(i)} \otimes \rho_B^{(i)},$$

then the state is **entangled**.
This is often verified using **mathematical criteria** such as the **Peres–Horodecki criterion**.

(b) Use the Partial Transpose Test (Peres–Horodecki Criterion):

1. Take the **partial transpose** of the density matrix with respect to one subsystem (say B):

$$\rho_{ij,kl}^{T_B} = \rho_{il,kj}$$

2. If any of the **eigenvalues of the partial transpose** are **negative**, then the state is **entangled**.
   - If all eigenvalues are **non-negative**, the state is **separable**.
   - This test is especially valid for **two-qubit** and **qubit-qutrit** systems.

Example:

For the **Bell state**,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

its density matrix is:

$$\rho_{\Phi^+} = \frac{1}{2}\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

The partial transpose of this matrix has a **negative eigenvalue**, confirming that the state is **entangled**.

4. Example: Mixed Entangled State (Werner State):

The **Werner state** is a common example used to test entanglement in mixed states:

$$\rho_W = p|\Psi^-\rangle\langle\Psi^-| + (1-p)\frac{I}{4}$$

where $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ is an entangled Bell state.

- When $p > \frac{1}{3}$, the state is **entangled**.
- When $p \leq \frac{1}{3}$, it becomes **separable**.

  This shows how entanglement can be quantitatively studied through the **density matrix and eigenvalue analysis**.